

Permutation Matrices as Generative Scores: Noise Models, Minimal Music, and Logic Programming

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We connect the algebraic structure of finite permutation matrices with Generative Logic by translating sequences of permutations into Prolog-based Simple Generative Logic Grammars (SGLGs). Permutation matrices are motivated as the “binary distillation” of unitary quantum evolution: they retain the norm-preserving and invertible character of unitary operators while restricting all matrix entries to $\{0, 1\}$. As a proof of concept, we use finite permutation matrices of rank $k = 8$ to generate structured sonic artifacts. By varying the choice of subsequent permutations, we explore three generative regimes: “white noise” (uncorrelated random permutations), “brown and pink noise” (random walks and fractal scaling on the Cayley graph), and “minimal music” trajectories (gradually reaching an arbitrary permutation of rank k through iterative transpositions in two-dimensional subspaces). The approach separates the algebraic progression from its sonic realization, highlighting the utility of permutation groups as formal resources for algorithmic composition and generative design.

Keywords: permutation matrices, unitary matrices, orthogonal matrices, logic programming, generative logic, algorithmic composition, noise models, minimal music, prolog, generative art, symmetric group, sound synthesis, quantum mechanics

I. INTRODUCTION

This paper connects algebraic combinatorics with algorithmic composition through the framework of Generative Logic [2, 3]. While earlier work demonstrated how finite partition logics can function as generative skeletons for visual artifacts [1], the present work investigates the *dynamic sequencing* of states. We ask whether sequences of finite permutation matrices can serve as structural engines for sound generation.

A key motivation comes from quantum mechanics. The time evolution of a closed quantum system is governed by *unitary* operators—complex matrices U satisfying $U^\dagger U = I$ —which preserve inner products and probabilities [4, 5]. Permutation matrices are the most elementary instances of unitary operators: they are unitary matrices whose entries are restricted to the binary set $\{0, 1\}$ rather than the full complex numbers. In this sense, a permutation matrix is a “binary distillation” of a unitary transformation, retaining the structural core—invertibility, norm preservation, and deterministic one-to-one mapping—while discarding all phase and amplitude freedom. Intermediate between permutations and full unitaries lie the *orthogonal* matrices (real entries, $O^\top O = I$), which preserve real inner products but allow continuous rotations and reflections. The resulting group-theoretic hierarchy,

$$S_k \subset O(k) \subset U(k), \quad (1)$$

means that every generative idea developed here for the finite group S_k has a natural continuous generalization to orthogonal and unitary trajectories.

In algorithmic composition, rule systems have long been used to constrain or generate sonic material—from Mozart’s

Musikalisches Würfelspiel to the stochastic music of Iannis Xenakis [16] and the algorithmic phase shifting of minimalist composers like Steve Reich and Philip Glass. A particularly close historical precedent is Josef Matthias Hauer’s *Zwölftonspiel* [21], which treats composition as the systematic permutation of twelve pitch classes within a trope constraint—an approach that we revisit and generalize in Sec. IX E.

Here, we formalize this generative practice using Prolog and finite symmetric groups. A sequence of permutation matrices operates on an initial vector of sonic events (e.g., an 8-note scale), rearranging it over time.

The aesthetic character of the generated sound depends entirely on the *trajectory* taken through the permutation space. We systematically construct three distinct categories of generative scores:

1. *White noise permutations*: unconstrained, purely random selections from S_k .
2. *Brown and pink noise permutations*: structurally correlated selections akin to random walks and fractal processes on the Cayley graph of S_k .
3. *Minimal music permutations*: deterministic, incremental phase-shifting processes that reach a global target via minimal rank-2 permutations (acting in two-dimensional subspaces).

By framing these concepts within a Simple Generative Logic Grammar (SGLG) [3], we achieve a clean separation between the mathematical structure (the permutation sequence) and the rendering map (the concrete synthesis of sound), following the three-layer architecture advocated in Ref. [1].

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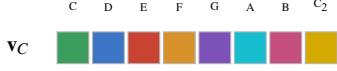


FIG. 1. The base vector $\mathbf{v}_C$ rendered as eight colored tiles. Color key used throughout: ■ C, ■ D, ■ E, ■ F, ■ G, ■ A, ■ B, ■ C_2 .

II. PERMUTATION MATRICES AS GENERATIVE SKELETONS

A. Formal structure

Let $\Omega_k = \{1, 2, \dots, k\}$ be a finite set. A *permutation* π of Ω_k is a bijection from the set to itself, belonging to the symmetric group S_k . This bijection is faithfully represented by a $k \times k$ *permutation matrix* P_σ with entries

$$(P_\sigma)_{ij} = \delta_{j, \sigma(i)}. \quad (2)$$

Every row and column contains exactly one 1; all other entries are 0. The set of all $k \times k$ permutation matrices forms a group under matrix multiplication isomorphic to S_k , of order $k!$ (for $k = 8$: $8! = 40320$).

The action of P_σ on a column vector $\mathbf{v} = (v_1, \dots, v_k)^T$ produces $\mathbf{v}' = P_\sigma \mathbf{v}$, reordering the entries of $\mathbf{v}$ according to σ . For artistic generation, $\mathbf{v}$ represents a palette of media events. We choose $k = 8$, corresponding to a diatonic octave:

$$\mathbf{v}_C = (C, D, E, F, G, A, B, C_2)^T. \quad (3)$$

The base vector is visualized as a row of colored tiles in Fig. 1; these eight colors recur in all subsequent figures.

B. Quantum-mechanical motivation: from unitary to permutation matrices

The choice of permutation matrices as primary generative entities is not arbitrary; it is rooted in the mathematical framework of quantum mechanics.

In quantum theory, the time evolution of a closed system with a k -dimensional state space is described by a $k \times k$ *unitary matrix* U , i.e., a matrix with complex entries satisfying [4, 5]

$$U^\dagger U = U U^\dagger = I, \quad (4)$$

where $U^\dagger$ denotes the conjugate transpose. Unitarity guarantees that the transformation is invertible, that it preserves the inner product $\langle \psi | \phi \rangle$, and hence that all probabilities and norms are conserved. The set of all such matrices forms the *unitary group* $U(k)$, a continuous Lie group of (real) dimension k^2 [6].

If one restricts attention to matrices with purely *real* entries, the unitarity condition $U^\dagger U = I$ reduces to orthogonality:

$$O^T O = O O^T = I, \quad O_{ij} \in \mathbb{R}. \quad (5)$$

These *orthogonal matrices* form the orthogonal group $O(k) \subset U(k)$, of dimension $k(k-1)/2$. Orthogonal transformations

preserve real inner products and include rotations and reflections in $\mathbb{R}^k$.

If one further restricts the matrix entries from the reals to the binary set $\{0, 1\}$, while still requiring that each row and each column contain exactly one nonzero entry, one obtains precisely the *permutation matrices*. Every permutation matrix is trivially orthogonal (and hence unitary), since its columns form an orthonormal set with respect to both the real and the complex inner product. But whereas $U(k)$ and $O(k)$ are continuous groups of infinite cardinality, the permutation matrices form the *finite* symmetric group S_k of order $k!$.

This yields the nested inclusion chain already stated in Eq. (1):

$$S_k \subset O(k) \subset U(k). \quad (6)$$

Table I summarizes the three levels. Moving upward in the hierarchy, one trades finiteness and combinatorial transparency for continuous phase freedom and richer interference effects. Moving downward, one distills the essential structural properties of unitary evolution—invertibility, norm preservation, one-to-one mapping of basis states—into a finite, combinatorial object.

This perspective clarifies why permutation matrices are a natural starting point for generative exploration. They are the *simplest nontrivial unitary operators*: they shuffle basis vectors without introducing any relative phases or superpositions. In quantum-information language, permutation matrices correspond to *classical reversible gates* acting on computational basis states [7]. They are exactly the unitary operators that map every computational basis state to another computational basis state, never creating superpositions.

The Birkhoff–von Neumann theorem [8] provides a further structural link: every *doubly stochastic* matrix (i.e., a nonnegative matrix whose rows and columns each sum to 1) can be written as a convex combination of permutation matrices. Since the modulus-squared entries of any unitary matrix, taken row-by-row or column-by-column, form a doubly stochastic matrix [9, 10], permutation matrices are the extreme points of the polytope that bounds all quantum-mechanically admissible probability tables.

For the generative framework of this paper, the key consequence is methodological. Every construction we develop for permutation sequences—white, brown, pink, and minimal-music regimes—can in principle be lifted to orthogonal or unitary trajectories, replacing the discrete Cayley graph of S_k by geodesics, Brownian motion, or fractional processes on $O(k)$ or $U(k)$ [12]. The permutation-matrix case is the “binary skeleton” of such continuous constructions: it captures

TABLE I. The hierarchy from unitary to permutation matrices. Each level preserves inner products and is invertible; the levels differ in the number field and the size of the resulting group.

Group	Entries	Dimension	Size
$U(k)$	$\mathbb{C}$	k^2	∞ (Lie)
$O(k)$	$\mathbb{R}$	$k(k-1)/2$	∞ (Lie)
S_k	$\{0, 1\}$	0 (discrete)	$k!$

the combinatorial essence while remaining finite, explicit, and implementable in a logic-programming environment.

A concrete illustration of the distillation is provided by the rank-2 case. A general 2×2 unitary matrix can be parameterized (up to a global phase) as

$$U_2(\theta, \phi, \lambda) = \begin{pmatrix} \cos \theta & -e^{i\lambda} \sin \theta \\ e^{i\phi} \sin \theta & e^{i(\phi+\lambda)} \cos \theta \end{pmatrix}. \quad (7)$$

Restricting to real entries ($\phi = \lambda = 0$) yields the 2×2 rotation (orthogonal) matrix

$$O_2(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (8)$$

Further restricting to $\{0, 1\}$ entries leaves only two possibilities: the identity ($\theta = 0$) and the swap

$$P_{\tau_{12}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (9)$$

which is the unique nontrivial 2×2 permutation matrix—our rank-2 transposition. The continuous rotation family (8) interpolates smoothly between identity and swap; the permutation matrix retains only the two endpoints. This is the sense in which permutations are the “binary distillation” of unitary (or orthogonal) transformations.

C. Transpositions and rank-2 permutations

Every $\sigma \in S_k$ can be written as a product of *transpositions* τ_{ij} , each swapping elements i and j . In matrix terms, $P_{\tau_{ij}}$ differs from the identity only in a 2×2 block acting on the span of the i and j basis vectors, exactly as in Eq. (9) (if i and j are adjacent, it is literally a contiguous 2×2 block). We call it a *rank-2 permutation matrix* because it acts nontrivially only on a two-dimensional subspace—the permutation-matrix analogue of a Givens rotation [11] in numerical linear algebra. Every $\sigma \in S_k$ can be decomposed into at most $k - 1$ transpositions; if one insists on *adjacent* transpositions $\tau_{i,i+1}$, the minimum number needed is the Kendall tau distance (number of inversions).

D. Concrete example

Consider $\pi = (1\ 3\ 5\ 7)(2\ 4\ 6\ 8) \in S_8$, which maps $1 \rightarrow 3$, $3 \rightarrow 5$, $5 \rightarrow 7$, $7 \rightarrow 1$, $2 \rightarrow 4$, $4 \rightarrow 6$, $6 \rightarrow 8$, $8 \rightarrow 2$. As a list: $\sigma = [3, 4, 5, 6, 7, 8, 1, 2]$. Figure 2(a) shows the 8×8 permutation matrix; panel (b) shows the result $P_\sigma \mathbf{v}_C$: the eight notes have been cyclically shifted in two interlocking quartets.

When a *sequence* of permutation matrices $P_1, P_2, \dots, P_N$ is applied iteratively to $\mathbf{v}_C$, it generates a score of length N : each “row” is a rearrangement of the eight base notes. How the sequence $\{P_t\}$ is selected dictates the “logic” of the musical composition.

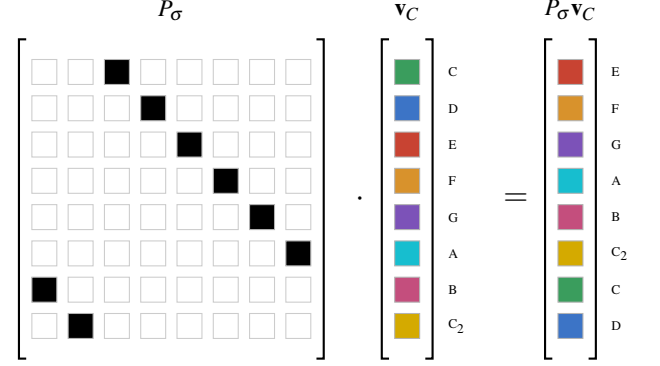


FIG. 2. The matrix equation $P_\sigma \mathbf{v}_C = \mathbf{v}'$ for $\sigma = (1\ 3\ 5\ 7)(2\ 4\ 6\ 8) \in S_8$. The 8×8 permutation matrix (black cells=1, white cells=0) acts on the color-coded base vector $\mathbf{v}_C = (C, D, E, F, G, A, B, C_2)^\top$, producing the output $\mathbf{v}' = (E, F, G, A, B, C_2, C, D)^\top$. Colors as in Fig. 1.



FIG. 3. **White noise**: six successive uniformly random permutations of rank 8, rendered as colored tile rows. Each row is a complete rearrangement of $\mathbf{v}_C$; no column shows temporal continuity. Colors as in Fig. 1.

III. WHITE NOISE PERMUTATIONS

A *white noise* sequence is generated by selecting each P_t independently and uniformly at random from S_k :

$$P_t \sim \mathcal{U}(S_k). \quad (10)$$

Algorithmically, a uniformly random permutation is produced by the Fisher–Yates shuffle [15].

In a musical context, this sequence lacks memory. Every step results in a complete reshuffling of the tonal material. Because the expected distance between P_t and P_{t+1} on the Cayley graph is near-maximal (a random permutation fixes on average only one element [15]), the resulting sound texture is chaotic and saturated.

In the quantum-mechanical analogy of Sec. II B, this regime corresponds to applying a sequence of independently Haar-random unitary matrices [13]—the unitary analogue of white noise—but distilled to the binary $\{0, 1\}$ skeleton of S_k .

Concrete example. Six successive random permutations of rank 8 and their color renderings are shown in Fig. 3. Note the complete lack of column-wise continuity: every color appears at unpredictable positions from row to row.



FIG. 4. **Brown noise**: a random walk on S_8 via adjacent transpositions, starting from the identity. At each step, the applied transposition $\tau_{i,i+1}$ is annotated on the right. Most columns show high continuity; changes propagate slowly.

IV. BROWN NOISE PERMUTATIONS

Brown noise (Brownian noise) in signal processing is generated by integrating white noise. In permutation space, this corresponds to a random walk on the Cayley graph of S_k with respect to adjacent transpositions [17]:

$$P_{t+1} = P_t \tau_{i_t, i_t+1}, \quad i_t \sim \text{Uniform}\{1, \dots, k-1\}. \quad (11)$$

Each step changes at most two voice assignments by one scale degree, so the spectral density of each voice’s pitch trajectory goes as $1/f^2$.

In the unitary hierarchy (6), the continuous analogue is Brownian motion on $O(k)$ or $U(k)$, generated by infinitesimal rotations drawn from the Lie algebra [12]. The permutation random walk is the binary skeleton of this process: the Lie-algebra generators (skew-symmetric matrices with continuous entries) are replaced by adjacent transpositions (the simplest generators of S_k).

Concrete example. Starting from the identity, a sample random walk of eight steps is shown in Fig. 4. At each step exactly one adjacent pair of tiles is swapped. The visual impression is one of slow drift: most columns retain their color across several time steps.

Aesthetically, a Brownian permutation sequence exhibits high structural continuity. Each generated row differs from the previous one by exactly two transposed elements. The listener perceives a slowly morphing textural field, where melody lines gradually evolve rather than randomly jump—reminiscent of the glissando textures in orchestral works by Xenakis [16].

The random walk on S_k is ergodic, with mixing time $\Theta(k^3 \log k)$ [17]; for $k = 8$ this is of order 10^3 steps.

V. PINK NOISE (FRACTAL) PERMUTATIONS

Pink noise ($1/f$ noise) sits between the chaotic unpredictability of white noise and the heavy correlation of brown noise, exhibiting fractal scale invariance [18, 19].

Direct generation of $1/f$ noise in a group-valued setting is nontrivial. We adopt a multi-scale superposition strategy inspired by the Voss–McCartney algorithm [19, 20]:



FIG. 5. **Pink noise**: eight steps of a Voss–McCartney permutation sequence with $K = 3$ adjacent-transposition registers. At $t = 4$ (arrow), all three registers update, causing a large rearrangement. Between major updates the sequence drifts smoothly.

1. Maintain $K = \lceil \log_2 k \rceil$ independent registers $R_0, \dots, R_{K-1}$, each holding an adjacent transposition.
2. At step t , re-randomize register m whenever bit m of the binary representation of t changes relative to $t-1$. Register 0 updates every step, register 1 every two steps, etc.
3. Compose: $P_t = R_{K-1} \circ \dots \circ R_1 \circ R_0$, applied to the identity.

In the unitary picture, the analogue would be a product of K independent Givens rotations at different time scales, yielding a fractal process on $O(k)$. The permutation version discretizes each factor to its two $\{0, 1\}$ -extremes: identity or swap.

Concrete example. Using $K = 3$ registers with adjacent transpositions, we traced eight steps explicitly. The register states and resulting permutations are:

t	R_0	R_1	R_2	Result
0	id	id	id	$[1, 2, 3, 4, 5, 6, 7, 8]$
1	$\tau_{3,4}$	id	id	$[1, 2, 4, 3, 5, 6, 7, 8]$
2	$\tau_{6,7}$	$\tau_{2,3}$	id	$[1, 3, 2, 4, 5, 7, 6, 8]$
3	$\tau_{1,2}$	$\tau_{2,3}$	id	$[2, 3, 1, 4, 5, 6, 7, 8]$
4	$\tau_{5,6}$	$\tau_{4,5}$	$\tau_{7,8}$	$[1, 2, 3, 6, 4, 5, 8, 7]$
5	$\tau_{2,3}$	$\tau_{4,5}$	$\tau_{7,8}$	$[1, 3, 2, 5, 4, 6, 8, 7]$
6	$\tau_{1,2}$	$\tau_{3,4}$	$\tau_{7,8}$	$[2, 1, 4, 3, 5, 6, 8, 7]$
7	$\tau_{6,7}$	$\tau_{3,4}$	$\tau_{7,8}$	$[1, 2, 4, 3, 5, 7, 8, 6]$

The color rendering appears in Fig. 5. At $t = 4$, all three registers update simultaneously, producing the largest visual discontinuity. Single-register updates ($t = 1, 3, 5, 7$) cause smaller changes. This multi-scale behavior yields a “musical” balance of repetition and surprise.

Table II compares the three stochastic regimes.

VI. MINIMAL MUSIC AND RANK-2 PERMUTATIONS

While the noise models rely on stochastic choices, *minimal music* (typified by Steve Reich’s *Piano Phase*) is characterized by strictly deterministic, gradual, and cumulative processes. We model this using purely algorithmic traversals of S_k .

TABLE II. Summary of the three noise regimes for permutation sequences ($k = 8$ voices) and their continuous analogues on $O(k)$ or $U(k)$.

Regime	Spectrum	$O(k)/U(k)$ analogue	Character
White	flat	Haar-random	chaotic
Brown	$1/f^2$	Brownian	slow drift
Pink	$1/f$	fractal	natural flow

Proposition 1. Any permutation matrix P of rank k can be reached from the identity I by a sequence of at most $k - 1$ transpositions acting in two-dimensional subspaces.

Proof. Standard: place elements into their correct positions sequentially; each placement requires at most one transposition. $\square$

In the language of the unitary hierarchy, this decomposition is the discrete analogue of factoring a unitary matrix into a product of Givens rotations [11] or two-level unitary matrices [7]. In the orthogonal case, any $O \in O(k)$ can be written as a product of at most $k(k - 1)/2$ Givens rotations; the permutation case collapses each continuous rotation to the binary choice of “swap or not.”

To generate a minimal music score, we *sonify the incremental journey* rather than jumping directly to the target:

$$P_0 = I, P_1 = T_1, P_2 = T_1 T_2, \dots, P_m = T_1 \dots T_m = P_{\text{target}}, \quad (12)$$

where each T_i is a rank-2 transposition. By repeating each state P_i multiple times (ostinato) before transitioning to P_{i+1} , the grammar produces the characteristic additive structure of minimalist composition.

Concrete example. Let the target be the “paired swap” permutation $\sigma_* = [2, 1, 4, 3, 6, 5, 8, 7]$, which exchanges every adjacent pair. A minimal decomposition uses four transpositions:

$$\begin{aligned} P_0 &= [1, 2, 3, 4, 5, 6, 7, 8], \\ P_1 &= [2, 1, 3, 4, 5, 6, 7, 8] \quad (\tau_{1,2}), \\ P_2 &= [2, 1, 4, 3, 5, 6, 7, 8] \quad (\tau_{3,4}), \\ P_3 &= [2, 1, 4, 3, 6, 5, 7, 8] \quad (\tau_{5,6}), \\ P_4 &= [2, 1, 4, 3, 6, 5, 8, 7] \quad (\tau_{7,8}). \end{aligned} \quad (13)$$

Figure 6(a) shows these five states as tile rows. Panel (b) shows the same path with each state repeated three times (ostinato), yielding the mesmerizing repetitive texture characteristic of minimal music.

Multiple targets can be concatenated into a cycle $I \rightarrow \sigma_1 \rightarrow \sigma_2 \rightarrow \dots \rightarrow \sigma_M \rightarrow I$, yielding a periodic score with hierarchical structure. The Cayley distance $d_C(\sigma) = k - c(\sigma)$ (where c counts cycles including fixed points) provides a natural measure of the musical *duration* for each transformation.

VII. COMPARATIVE OVERVIEW

To explicitly synthesize the generative behaviors outlined in the preceding sections, Figure 7 juxtaposes all four regimes

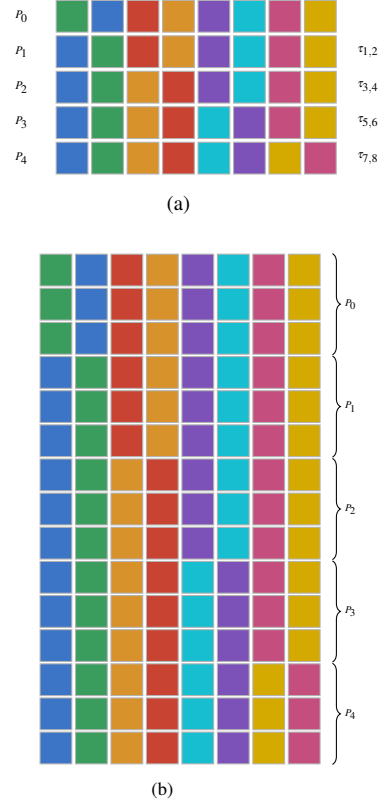


FIG. 6. **Minimal music:** path from identity to the paired-swap permutation $[2, 1, 4, 3, 6, 5, 8, 7]$ via four transpositions. (a) Bare path. (b) Ostinato rendering ($3 \times$ repetition per state).

in a single display. The visual differences across the panels translate directly into their corresponding acoustic characteristics: the complete lack of vertical coherence in the white noise panel (a) yields a chaotic, saturated texture; the subtle, single-swap transitions in the brown noise panel (b) create a slow, drifting musical evolution; the Voss–McCartney multi-scale updates in the pink noise panel (c) balance recognizable structural continuity with sudden shifts; and the rigid, cumulative rank-2 transformations in the minimal music panel (d) visually capture the mechanical phase-shifting inherent in deterministic algorithms.

VIII. PROLOG REALIZATION

We implement all four regimes in SWI-Prolog [14], separating the structural layer (permutation generation) from the rendering layer (MIDI pitch assignment), following the SGLG framework of [3].

The general SGLG for an N -step score is

$$S \rightarrow \text{row}_1 \text{ row}_2 \dots \text{row}_N, \quad (14)$$

$$\text{row}_j \rightarrow \text{render}(P_j \mathbf{v}_C) \text{ br}. \quad (15)$$

Listing 1. Core predicates: permutation representation, Fisher–Yates shuffle, and swap.

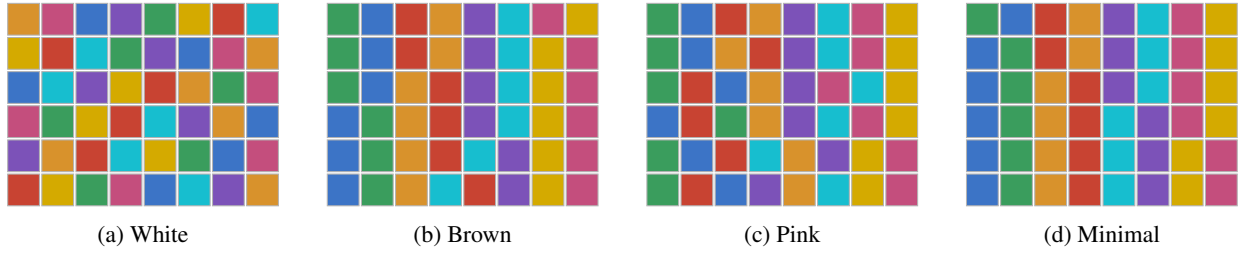


FIG. 7. Side-by-side comparison of six time steps under the four regimes (all $k = 8$). (a) White noise: no column coherence. (b) Brown noise: high local continuity. (c) Pink noise: multi-scale changes. (d) Minimal music: deterministic cumulative swaps. Colors as in Fig. 1.

```

:- use_module(library(random)).
:- use_module(library(lists)).

identity(N, P) :- numlist(1, N, P).

% Fisher-Yates shuffle (white noise)
random_perm(N, P) :-
    identity(N, I),
    fisher_yates(I, P).
fisher_yates([], []).
fisher_yates(L, [H|T]) :-
    length(L, Len),
    random_between(1, Len, K),
    nth1(K, L, H),
    select(H, L, Rest),
    fisher_yates(Rest, T).

% Adjacent transposition
adj_swap(K, Pin, Pout) :-
    nth1(K, Pin, A), K1 is K+1,
    nth1(K1, Pin, B),
    set_nth(K, Pin, B, Tmp),
    set_nth(K1, Tmp, A, Pout).
set_nth(1, [_|T], X, [X|T]).
set_nth(N, [H|T], X, [H|R]) :-
    N>1, N1 is N-1,
    set_nth(N1, T, X, R).

% General swap
swap(I, J, Pin, Pout) :-
    nth1(I, Pin, A), nth1(J, Pin, B),
    set_nth(I, Pin, B, Tmp),
    set_nth(J, Tmp, A, Pout).

```

Listing 2. Brown, pink, and minimal-music generators.

```

% Brown noise: random walk
brown_step(N, Pin, Pout) :-
    Max is N-1,
    random_between(1, Max, K),
    adj_swap(K, Pin, Pout).
brown_seq(_, P, 0, [P]).
brown_seq(N, P, S, [P|R]) :-
    S>0, brown_step(N, P, P1),
    S1 is S-1, brown_seq(N, P1, S1, R).

% Pink noise (Voss-McCartney)
pink_step(T, N, Rs0, Rs1, Perm) :-
    update_regs(T, 0, N, Rs0, Rs1),
    identity(N, Id),
    compose_regs(Rs1, Id, Perm).

```

```

update_regs(_, _, _, [], []).
update_regs(T, K, N, [R|Rs], [R1|Rs1]) :-
    B is (T>>K)/1,
    P is ((T-1)>>K)/1,
    (B=P -> rand_adj(N, R1); R1=R),
    K1 is K+1,
    update_regs(T, K1, N, Rs, Rs1).
rand_adj(N, swap(A, B)) :-
    Max is N-1,
    random_between(1, Max, A), B is A+1.
compose_regs([], P, P).
compose_regs([swap(A, B)|Rs], Pin, Po) :-
    swap(A, B, Pin, P1),
    compose_regs(Rs, P1, Po).

% Minimal music: decompose into swaps
minimal_path(P, P, []).
minimal_path(Cur, Tgt, [swap(I, J)|Ss]) :-
    Cur = Tgt,
    first_diff(Cur, Tgt, 1, I),
    nth1(I, Tgt, V),
    nth1(J, Cur, V),
    swap(I, J, Cur, Next),
    minimal_path(Next, Tgt, Ss).
first_diff([H|_], [H2|_], K, K) :- H=H2, !.
first_diff([_|A], [_|B], K, R) :-
    K1 is K+1, first_diff(A, B, K1, R).

```

Listing 3. Rendering layer and DCG output.

```

% Map to MIDI pitch (C major scale)
pitch(Deg, Midi) :-
    Scale=[0,2,4,5,7,9,11,12],
    D1 is Deg-1, nth0(D1, Scale, Off),
    Midi is 60 + Off.

% DCG rendering
render([]) --> [].
render([H|T]) --> note(H), render(T).
note(1)-->[c4]. note(2)-->[d4].
note(3)-->[e4]. note(4)-->[f4].
note(5)-->[g4]. note(6)-->[a4].
note(7)-->[b4]. note(8)-->[c5].
br --> [rest].

% Main entry points
play_white(N, Steps) :-
    forall(between(1, Steps, _),
        (random_perm(N, P),
         phrase(render(P), Out),
         writeln(Out))).

play_minimal(N, Target) :-

```

```

identity(N, Id),
minimal_path(Id, Target, Swaps),
replay(Id, Swaps).
replay(P, []) :-
    phrase(render(P), Out), writeln(Out).
replay(P, [swap(I, J) | Ss]) :-
    phrase((render(P), br), Out),
    writeln(Out),
    swap(I, J, P, P1), replay(P1, Ss).

```

The output of `play_minimal(8, [2, 1, 4, 3, 6, 5, 8, 7])` produces exactly the five-row terminal sequence depicted in Fig. 6(a). When this list of terminal symbols is passed to a MIDI engine or via OSC to a synthesizer, the algebraic path becomes an audible score.

IX. DISCUSSION

A. The unitary-permutation bridge

The hierarchy $S_k \subset O(k) \subset U(k)$ is not merely a mathematical observation; it has direct generative consequences. Every regime defined in this paper—white, brown, pink, minimal—has a continuous counterpart on $O(k)$ or $U(k)$. White noise on S_k corresponds to Haar-random sampling on $U(k)$ [13]. Brown noise corresponds to Brownian motion on the Lie group [12]. The minimal-music decomposition into transpositions corresponds to factoring a unitary into Givens rotations or two-level unitaries [7, 11].

Working with permutations rather than continuous matrices offers two practical advantages. First, the output at each time step is a *deterministic reassignment* of the k voices—no superpositions, no fractional amplitudes—which maps directly to the combinatorial constraints of traditional polyphonic writing. Second, the finite group S_k is exactly enumerable and implementable in a logic-programming language without floating-point arithmetic.

At the same time, the unitary perspective suggests natural extensions. One could interpolate between two permutation matrices via a geodesic on $O(k)$ or $U(k)$, producing a continuous “crossfade” between two discrete voice assignments. Such interpolations would introduce the relative phases and superpositions characteristic of quantum mechanics into the generative framework.

B. Structure as constraint

In classical generative art, random noise is often applied directly to synthesis parameters. In our method, randomness operates on the *syntactic skeleton*—the permutation matrix. The result is a composition that never violates the set constraints: no matter how chaotic a white-noise permutation becomes, the output row contains exactly all k notes without duplication. The algebraic properties of the permutation matrix act as an unbreakable conservation law for the musical material.

C. Perceptual offerings and time

When we compare the partition-logic visual artifacts of Ref. [1] with the permutation-based musical scores generated here, the crucial variable is *time*. A visual grammar presents its entire derivation simultaneously; a musical grammar is a *temporal perceptual offering*. The brown-noise sequences afford acts of auditory tracking—the listener can hear the rank-2 transpositions bubbling through the sequence. The minimal-music trajectories afford a sense of geometric convergence, building tension as the vector is slowly sorted into the target permutation.

D. Generalization

The construction generalizes immediately from $k = 8$ to arbitrary rank. Larger k yields denser pitch fields and longer minimal paths; smaller k produces sparser, more audibly transparent textures. The same permutation sequence can be rendered as pitch assignments, rhythmic patterns, spatial positions, or timbral interpolations by changing only the rendering predicates.

E. Connection to Hauer’s *Zwölftonspiel*

The permutation-based framework developed in this paper has a significant historical precedent in the work of the Austrian composer Josef Matthias Hauer (1883–1959) [21–23]. Hauer developed his twelve-tone method independently of—and slightly prior to—Arnold Schoenberg [25, 26], but his approach is arguably more purely permutational and therefore closer in spirit to the present work.

1. Hauer’s system as permutation music

In Hauer’s *Zwölftonspiel* (literally: twelve-tone game or play), the chromatic aggregate $\Omega_{12} = \{0, 1, \dots, 11\}$ of twelve pitch classes is the primary material. A twelve-tone row is a permutation $\sigma \in S_{12}$ of this set: every pitch class appears exactly once, and the compositional act consists in choosing and ordering these permutations. This is precisely the framework of the present paper, specialized to $k = 12$ and the chromatic rather than diatonic scale.

The constraint that every row is a *complete* permutation of all k pitch classes—no omissions, no repetitions—functions in both Hauer’s system and in our framework as an “unbreakable conservation law” for the musical material. In our language (Sec. IX), the doubly stochastic property of the permutation matrix guarantees this conservation at every time step.

2. Tropes as equivalence classes on S_{12}

Hauer’s most distinctive contribution is the concept of the *Trope* (trope) [21, 23]. He partitions the twelve pitch classes

TABLE III. Comparison of Hauer’s *Zwölftonspiel*, Schoenberg’s twelve-tone method, and the present framework. All three operate on a symmetric group S_k but differ in the constraints and traversal strategies applied.

	Hauer	Schoenberg	This paper
k	12	12	arbitrary
Constraint	trope	fixed row	none / adj.
Row forms	$(6!)^2$	≤ 48	$k!$
Traversal	free	deterministic	white/brown/ pink/minimal
Transformation group	$\mathbb{Z}_{12}$ (trope level)	$D_{12} \times \mathbb{Z}_2$	S_k

into two complementary *hexachords* (sets of six), disregarding the internal ordering within each half. Two such partitions are considered equivalent if one can be obtained from the other by transposition (cyclic shift in $\mathbb{Z}_{12}$).

In group-theoretic terms, a trope is an orbit of the action of $\mathbb{Z}_{12}$ on the set of unordered complementary hexachord pairs:

$$\mathcal{T} = \left\{ \{H, \Omega_{12} \setminus H\} \mid H \subset \Omega_{12}, |H| = 6 \right\} / \mathbb{Z}_{12}. \quad (16)$$

Hauer identified exactly 44 such trope classes [22–24], a count that can be verified by Burnside’s lemma [27].

Within each trope, the internal ordering of the two hexachords is free—that is, any element of $S_6 \times S_6$ may be applied independently to the two halves. The trope therefore acts as a *macro-structural constraint* (which six pitches go together), while the micro-structure (the ordering within each hexachord) provides a space of $(6!)^2 = 518400$ possible realizations per trope.

This two-level architecture—a coarse partition into tropes, followed by free permutation within each half—is a precursor of the three-layer separation (formal skeleton, generative grammar, rendering map) advocated in Refs. [1, 3] and employed throughout the present paper.

3. Hauer versus Schoenberg

It is instructive to contrast Hauer’s approach with Schoenberg’s better-known twelve-tone technique [25]. Schoenberg fixes a single *tone row* (a specific permutation $\sigma_0 \in S_{12}$) and derives all subsequent material from it by four operations: identity, retrograde R , inversion I , and retrograde inversion RI , each combined with the twelve transpositions T_n . This yields a group of exactly 48 row forms—a small subgroup of S_{12} .

Hauer, by contrast, does not privilege a single row. His *Spiel* (game) character allows free navigation within the vast space of $12! = 479001600$ permutations, subject only to the trope constraint. In the terminology of this paper, Schoenberg’s method is a *minimal-music regime* (a small, deterministic set of transformations applied to a fixed starting point), whereas Hauer’s method is closer to a *constrained random walk* on S_{12} , with the trope acting as a hyperplane constraint that restricts the walk to a subset of the Cayley graph.

Table III summarizes the comparison.

4. Hauer’s tropes and noise regimes

The connection between Hauer’s tropes and our noise regimes is suggestive. A *white-noise Zwölftonspiel* would draw each successive row uniformly at random from the 518400 permutations compatible with a given trope—a constrained version of our white-noise regime (Sec. III). A *brown-noise Zwölftonspiel* would perform a random walk on the Cayley graph of $S_6 \times S_6$ within the trope, applying adjacent transpositions independently to the two hexachords—producing the slowly drifting texture of Sec. IV, but with the added structural feature that the hexachord boundary is never crossed. A *minimal-music Zwölftonspiel* would reach a target permutation within the trope by a sequence of at most 10 transpositions (at most 5 per hexachord), yielding a Reich-like phasing process on the chromatic aggregate.

These constructions suggest that Hauer’s trope system can be naturally embedded within our generative framework by restricting the permutation group from S_{12} to $S_6 \times S_6$ (or, more precisely, to the coset of $S_6 \times S_6$ in S_{12} defined by the trope).

5. The Spiel as generative grammar

Finally, Hauer’s deliberate use of the word *Spiel* (game, play) for his compositional method resonates with the generative-logic perspective of Jendreiko’s SGLGs [2, 3]. In both cases, the compositional act is framed not as the expression of subjective intention but as the disciplined exploration of a combinatorial space defined by explicit rules. The composer does not “write” the music; rather, the composer *sets up the game* and lets the rules generate the material. This is precisely the ethos of Generative Logic applied to sonic production.

Hauer’s work thus appears, in retrospect, as an early instance of permutation-based generative composition—a practice that the present paper formalizes, generalizes, and implements in executable logic-programming form.

X. CONCLUSION

We have demonstrated that sequences of finite permutation matrices—the binary distillation of unitary quantum evolution—serve as powerful structural engines for generative logic programming and algorithmic composition. The group-theoretic hierarchy $S_k \subset O(k) \subset U(k)$ situates our constructions within the broader landscape of quantum-mechanical transformations: permutation matrices retain the essential properties of unitarity (invertibility, norm preservation, one-to-one basis mapping) while restricting all entries to $\{0, 1\}$.

By mapping the identity matrix to a base musical vector, we translate paths on the Cayley graph of S_k into perceptible sonic experiences. We formulated three stochastic regimes—white, brown, and pink noise—and one deterministic regime inspired by minimal music, each with a well-defined continuous analogue on $O(k)$ or $U(k)$.

We have also shown that the framework naturally subsumes Hauer's *Zwölftonspiel* as the special case $k = 12$ with a trope constraint restricting the permutation group to $S_6 \times S_6$, thereby connecting early twentieth-century twelve-tone practice to contemporary logic programming and noise-theoretic models.

Through Prolog-based SGLGs, the matrix arithmetic is formalized as an executable logic grammar, explicitly isolating the mathematical skeleton from the acoustic rendering. The permutation matrix is transformed from a dry algebraic operator into an expressive, generative score.

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